A3-PS019-TC316

Title: Privacy Intelligence Engine
Personal-Data Discovery & Privacy Evidence Inventory

Team: Cipher Shifter

Problem Description

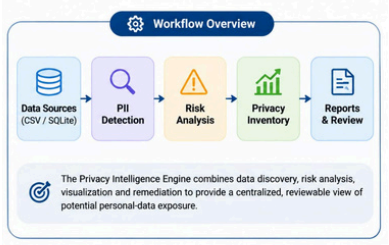
Abstract:
A Streamlit-based application that discovers potential PII in CSV and SQLite data, prioritizes privacy risks, and provides evidence inventory, remediation guidance, and reporting.

Problem:

- Manual PII discovery is time-consuming.
- Personal data may be scattered across sources.
- Privacy risks are difficult to prioritize.
- Centralized evidence and reporting are needed.

Background

- Personal data is stored across multiple structured sources.
- Identifying sensitive fields and assessing potential exposure is important for privacy governance.
- The proposed system combines automated discovery with risk visualization and human-reviewed remediation.



Literature & RFC Review

- **DPDP Act, 2023:** Indian digital personal-data protection framework.
- **NIST Privacy Framework:** Privacy risk management.
- **ISO/IEC 27701:** Privacy information management.
- **Existing approaches:** Manual review and pattern-based detection.

Research gap: Unified discovery, prioritization, and evidence reporting.

Proposed Solution

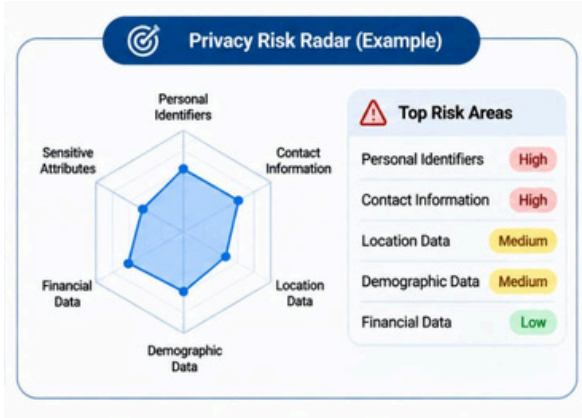
A centralized privacy tool supporting:

- CSV/SQLite data discovery
- PII candidate identification
- Risk prioritization and exposure visualization
- Privacy posture assessment
- Remediation recommendations
- Benchmark evaluation and report export



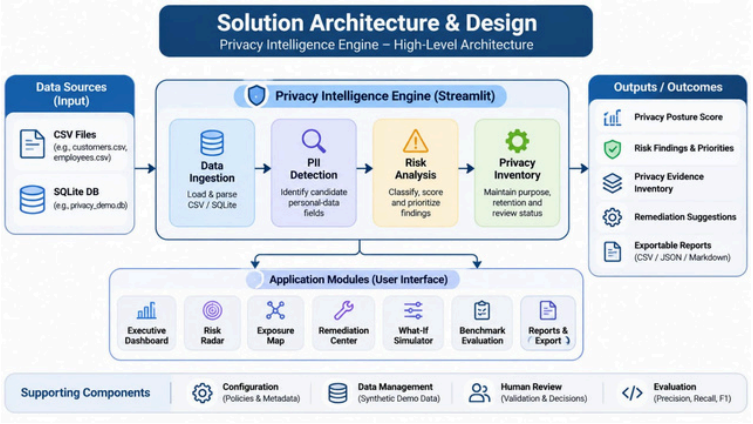
Optimization Proposed

Traditional Approach	Proposed System
Manual Inspection	Automated Scanning
Scattered Findings	Centralised Inventory
Manual Proioritizing	Risk Radar
Manual Reporting	Report Export



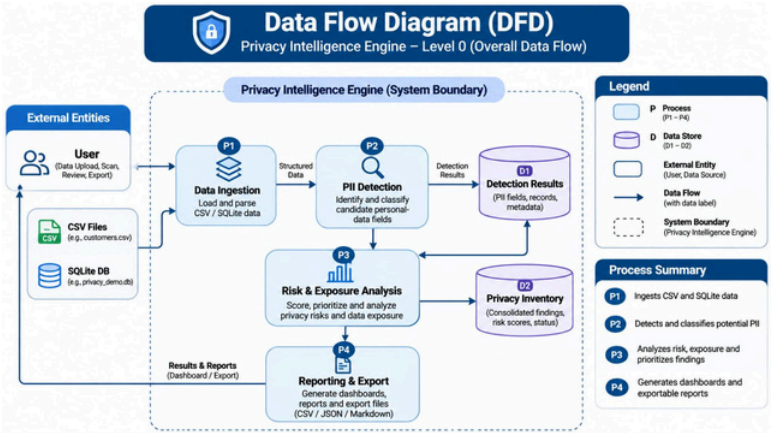
Solution Architecture & Design

User → Streamlit UI → CSV/SQLite Ingestion → PII Detection → Risk Analysis → Dashboard & Reports



Data Flow Diagram

CSV/SQLite → Ingestion → PII Detection → Evidence Inventory → Risk Analysis → Visualization/Export



Deployment Timeline

Phase	Activity
1	Requirement And Design
2	Data Integration And PII Detection
3	Risk Analysis And Visuakisation
4	Testing And Evaluation
5	UI Refinement Documentation

Bibliography I

1. Digital Personal Data Protection Act, 2023.
2. NIST Privacy Framework.
3. ISO/IEC 27701.
4. Streamlit Documentation.
5. Python/Pandas/Plotly documentation, as applicable.

THANK YOU

Privacy Intelligence Engine
Making Personal-Data Discovery More Visible and Actionable

PS-019 | Team A3-PS019-TC316